

A spatial optimization model for resource allocation for wildfire suppression and resident evacuation

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ABSTRACT

Wildland-urban interface wildfires have been a significant threat in many countries. This paper presents an integer two-stage stochastic goal programming model for comprehensive, efficient response to a wildfire including firefighting resource allocation and resident evacuation. In contrast to other natural disasters, the progression of wildfires depends on not only the probabilistic fire-spread scenarios but also decisions made during firefighting. The proposed model optimizes the resource preparations before the fire starts and resource allocation decisions during the fire event. The model takes into account different wildfire-spread scenarios and their impact on high-risk areas. The two objectives considered are minimizing the total cost of operations and property loss and minimizing the number of people at risk to be evacuated. A case study based on Santa Clara County in California, United States of America, is presented to demonstrate the model performance. Numerical experiments show that this model works well to find solutions by considering a trade-off between two objectives; and varying cell size based on scenarios, reduces problem dimension and improves solution time.

1. Introduction

1.1. Background

Wildfires in the urban-wildland interface (WUI) have been severe threats to human beings and wildlife. Wildfire management is a challenging and ongoing problem. Despite considerable efforts invested in preventing wildfires, disastrous wildfires still occur every year worldwide. Over the past few years, the WUI wildfires have become greater threats in the United States, Australia, and Portugal. In 2017, wildfires in both North and South California led to losses of more than 10 billion US dollars (Yan, 2017). According to a report from California Department of Forestry and Fire Protection (CAL FIRE), the October 2017 Northern California wildfires, including more than 250 wildfires, resulted in 43 fatalities and widespread evacuations of more than 100,000 people (CAL FIRE, 2017). These fires burned over 245,000 acres, and at least 11,000 firefighters participated in fire containment. The December 2017 Thomas Fire in Southern California charred 281,893 acres, destroyed more than 1300 structures, and forced over 100,000 people to evacuate (CAL FIRE, 2018c). As of December 2018, the Camp Fire, which started on November 8th, 2018 in North California and caused 86 civilian fatalities, was the deadliest wildfire in modern California history (CAL FIRE, 2018a). All these fires were

typical WUI wildfires. Unlike wildfires in wildland, WUI wildfires threaten both the ecosystem and the communities.

According to the wildfire paradox, most wildfires could be controlled by a variety of regular suppression measures, while the remaining wildfires could become disastrous (Calkin, Cohen, Finney, & Thompson, 2014). It is estimated that the largest 1% of the wildfires are responsible for 83–96% of the burned areas (Calkin, Gebert, Jones, & Neilson, 2005; Strauss, Bednar, & Mees, 1989). Since it is not possible to eliminate the occurrence of wildfires, developing effective strategies to prevent the fires from becoming catastrophic is essential. The critical challenge is the uncertainty in fire-spread speed and direction. Even though the initial size and spread speed become available after a fire breaks out, making a detailed containment plan is still challenging. In contrast to other natural disasters which usually happen at a particular time point and are followed by relief efforts, wildfires may keep expanding unless proper and sufficient intervention plans are put into action. These actions involve both containment efforts and evacuation of residents. In addition, resources are often limited to execute all of the necessary actions. Thus, the prioritization of efforts in different wildfire regions becomes crucial.

Despite the fact that the suppression costs have exceeded 1 billion dollars annually, there is limited to no research on the effectiveness of suppression efforts and modeling resource allocation decisions for fire

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containment (Finney, Grenfell, & McHugh, 2009; Pacheco et al., 2015). The main objective of this paper is to propose and solve a mathematical model for the problem of wildfire preparation and response in a WUI area, which minimizes number of residents at risk, property losses, and fire containment cost.

The challenges of modeling wildfire management problem are led by the uncertainty of fire size and the dynamic process of fire containment. Due to several factors, wildfires may spread with an unpredictable speed towards different directions. A reasonable mathematical model should consider possible fire scenarios with different rates and directions. However, even if the initial condition of the fire becomes known right after a fire breaks out, firefighters' activities will keep affecting the fire-spread, which makes the modeling efforts more complicated as the data may present fire-spread in the presence of firefighting interventions. In addition, in some rapid-fire cases, fire containment resources may be insufficient for initial response, which leads to another challenge of making plans for fire containment.

In this paper, a mathematical optimization model is developed for management of WUI wildfires, which first, minimizes the number of people at risk in the high-risk areas and second, minimizes the total cost of fire containment and property losses. The model optimizes the resource allocation decisions given limited capacity and availability while taking into account possible wildfire-spread scenarios. Scenarios in the model are based on fire-spread direction and speed, which are determined by different wind directions and strength. In the proposed model, number of decision variables is kept constant in each scenario by developing a new approach for geographic area representation, so that solutions are found to highly comprehensive realistic WUI wildfire problems with varying area sites in different scenarios. Solutions to the model, which show the trade-off between objectives, can help decision makers to make choices based on the actual situations they might be facing.

The paper is organized as follows: Section 1.1 introduces the background of wildfire management problem. Section 1.2 includes relevant literature on wildfire preparedness and response operations. Section 2.1 presents the model assumptions, data sources, the stochastic goal programming model and a simple computational experiment. Section 3 demonstrates a case study from Santa Clara County. In Section 4, conclusions are summarized.

1.2. Literature review

Previous modeling efforts have considered different aspects of wildfire management, including developing strategies for fire prevention or mitigating potential effects, finding action plans during fire suppression while minimizing costs, or planning for evacuation.

One of the main factors that affect the spreading of a wildfire is the nature of the land. In order to reduce the risks of potential wildfires, preventive actions, often called "fuel management," such as removing live and dead vegetation are utilized. Minas, Hearne, and Martell presented several models for fuel management problems to reduce wildfire effects (Minas, Hearne, & Martell, 2014) and prepare for fire suppression (Minas, Hearne, & Martell, 2015). Rashidi, Medal, Gordon, Grala, and Varner proposed a model to evaluate the worst-case scenario in a specific wildfire (Rashidi, Medal, Gordon, Grala, & Varner, 2017). These models take into account terrains and preventative actions. However, models that take into account resource preparation, show the process of fire-spread, and consider the effects of the response process are still needed.

Most of the operations management literature includes operations research models for optimal resource allocation with the objective of minimizing costs. These models use linear programming or stochastic programming to optimize resource allocation decisions. MacLellan and Martell built an integer programming model to determine the initial locations (home bases) and deployment strategy for airtankers while minimizing the average annual cost of satisfying the demand for

airtankers during firefighting season (MacLellan & Martell, 1996). Similarly, Dimopoulou and Giannikos implemented a maximal coverage location model to optimize the locations of fire fighting vehicles to cover the demand areas (Dimopoulou & Giannikos, 2002). In a follow-up study, they provided an integrated framework for fire forest fire control that consists of a GIS system, fire prediction model, a maximal coverage location model and a simulation model to test different scenarios and strategies (Dimopoulou & Giannikos, 2004). For a single fire, Donovan and Rideout built an integer programming model to minimize the sum of suppression cost, preparation cost, and damage cost by optimizing resource allocation (Donovan & Rideout, 2003). For multiple fires on the same day, Hu and Ntamo presented a stochastic optimization model to manage fire initial attack response and a simulation model for the suppression process (Hu & Ntamo, 2009). Based on historical data, Arrubla, Ntamo, and Stripling set up an integer programming model to plan initial response actions (Arrubla, Ntamo, & Stripling, 2014). Pacheco and Claro suggested that whether the weather is rainy or dry also produces uncertainties to wildfire. Their model considered effects of weather variability on overall fire season and ignition probability, and prepared suppression resources for the worst wildfire case (Pacheco & Claro, 2018).

In addition to minimizing costs, fire management involves other objectives such as maximizing the remaining land value (Pappis & Rachaniotis, 2010), preserving wildlife and the habitat, health, air and water quality (Venn & Calkin, 2011). Even though finding the actions that minimize operational costs is important, saving lives is more critical during a wildfire event. During WUI fires, saving lives within a minimal amount of time is a great challenge. Cova, Dennison, and Kim (2005) presented methods for efficient evacuation such as setting evacuation trigger criteria based on a fire-spread model using a geographical information system (Cova et al., 2005). Li, Cova, and Dennison (2015) also proposed setting evacuation alerts based on trigger modeling (Li et al., 2015). These models mainly focus on initial response to a fire.

The abovementioned objectives in wildfire management often conflicts with each other. Thus, multi-objective optimization models to find Pareto-optimal solutions, i.e. solutions in which none of the objective values can be improved without degrading others, are sought and the trade-offs between different objectives are examined. Calkin, Hummel, and Agee used goal-programming to study the trade-offs between fire threat reduction and ecosystem preservation for fire preparedness (Calkin, Hummel, & Agee, 2005). Another objective that is relevant in urban-wildland fires is to prevent residential losses. Firefighting and resident evacuation influence each other. They are usually planned by the same fire department or emergency operations center as a multi-goal task. Achieving multiple goals has been a common objective not only in wildfire management, but also in several different kinds of disaster response. Zhan and Liu built a multi-objective stochastic programming model that minimizes both travel time and proportion of unmet demand to disaster response (Zhan & Liu, 2011). This model handles disasters occurring in a specific location and timing such as earthquakes and hurricanes. Many disaster preparedness and relief distribution models have similarities with fire management models. For instance, Tzeng, Cheng, and Huang, in an earthquake preparedness and relief context, proposed an optimization model for location and distribution of commodities while simultaneously optimizing the objectives of minimizing cost, minimizing travel time, and maximizing the minimal satisfaction (Tzeng, Cheng, & Huang, 2007). Mete and Zabinsky built a stochastic optimization model for medical supply location and distribution for disaster management (Mete & Zabinsky, 2010). Murali, Ordóñez, and Dessouky presented a location allocation model for large scale bi-terrorism attack (Murali, Ordóñez, & Dessouky, 2012). Yi and Özdamar proposed a multi-commodity network flow model for distributing necessary material, personnel or equipment to emergency areas while minimizing the delay in arrival times (Yi & Özdamar, 2007). For more detailed reviews of disaster preparedness and relief

models in operations management literature, the readers are referred to Balcik and Beamon (2008), Caunhye, Nie, and Pokharel (2012), and Hoyos, Morales, and Akhavan-Tabatabaei (2015). Most of the disaster management models optimize pre-disaster preparedness efforts before a random disaster occurs, and post-disaster relief efforts once a disaster happens and its impacts are already observed. However, wildfires differ from other disasters such as earthquakes and hurricanes, as wildfires keep spreading if no actions are taken, and firefighting interventions may affect the fire-spread significantly.

Most of the wildfires are controlled by local resources (Center, 2004), but there are still a few disastrous wildfires that require the cooperation of teams from different counties, states or even countries. Usually, local fire-fighting resources would respond to suppress the initial fire attack, and more nonlocal resources would become available for the response if a fire keeps spreading. Different teams from various areas usually have agreements and plans. For example, the California Fire Service and Rescue Emergency Mutual Aid Plan was adopted to provide a systematic and efficient response when a disaster happens (California Governor's Office of Emergency Services, 2014).

This paper presents a two-stage stochastic integer goal programming model, which focuses on optimizing the resource allocation considering different wildfire scenarios while minimizing both the number of people in danger and the total cost of fire suppression and land losses. In this model, decisions of containment and evacuation are made simultaneously so that they will be more efficient and better reflect reality. As urban areas continue expanding and fire risks in different seasons differ, decisions of on-duty resources scheduled in each fire station need to be made in advance to prepare for possible wildfires. Unlike the models mentioned above, population and property density are both critical factors in this model, which affect both containment and evacuation decisions.

2. Mathematical model

2.1. Model assumptions and data sources

Fire suppression is a multi-stage process as different types of resources (local or nonlocal) become available at different times if a fire is not contained immediately. The model presented in this paper formulates this multi-stage process as a two-stage stochastic program. Fig. 1 shows the stages of wildfire containment process considered in this model. The decisions in the first stage involve hiring of additional on-duty resources for every station for the fire season that is under consideration. Once a fire starts, an assumption is made that the uncertainties about fire-spread speed and direction will be revealed. The uncertainty about fire-spread speed and direction is incorporated in the model as scenarios with varying probability of occurrences. In the second stage, a detailed fire suppression plan is made for each specific

fire scenario. Decisions in this stage include identifying high-risk areas for evacuation and allocating resources by their availability. Resource availability means some of the resources cannot join fire suppression until a specific period. For instance, resources outside of California cannot be allocated to contain the fire in California until the declaration of a state of emergency, which usually happens 24 h after fire initiation (CAL FIRE, 2004). Since the operation cost of on-duty resources is lower than any other resources, on-duty resources will always be sent first if a fire happens. Similarly, nonlocal resources will not be considered unless all the available local resources are utilized, or nonlocal resources are cheaper in some particular condition.

The model assumes that all of the resource allocation decisions are based on the assessment of the fire condition so that more resources are allocated to fires that are substantial. Based on the fire condition, geographical areas are determined as high-risk, completely burned, or treated (under control). By keeping track of fire conditions in each area, resource allocation and evaluation of property losses are determined.

Many factors contribute to fire initiation and spread such as fuel, wind, humidity, and season. This model does not consider fuel management efforts for fire prevention, since fuel management decisions and actions are made well ahead of fire season and utilize different resources. Thus, when analyzing a specific location with certain fuel in a specific season, it is assumed that the only uncertainty is caused by the wind and humidity. In this model, scenarios are generated considering humidity, different wind directions, and different wind strengths. Fire-spread direction is estimated based on wind direction, and fire-spread speed is estimated based on wind strength and humidity. In addition, for a specific season, the probability of each scenario is estimated by analyzing historical data from a specific region with known fuel type.

This model demonstrated spatial fire-spread process and detailed allocation plans by expressing geographical areas as cells. Cell sizes are associated with the fire-spread rate and length of each period. In other words, the cell size in a fast fire scenario is larger than the cell size in a slow fire scenario. Varying cell sizes keep the number of decision variables constant in different scenarios with different spread speed, which as a result enables us to solve problems that involve very large geographical areas. For different cell sizes in different fire-spread scenarios, cell population density and cell land value are generated using publicly available data sets. Each cell in the absence of a fire event is considered as a no-risk or safe cell. At the beginning of each possible fire, the initial conditions of cells including the fire-starting point and adjacent cells to the fire-starting point are updated. These initial cell conditions are determined as either high-risk, which means the cell is about to be affected by fire-spread or has been partially on fire, or burnt, which means the cell has been fully on fire. In the following periods, suppression efforts based on resource allocation decisions will affect cells' conditions and result in an update.

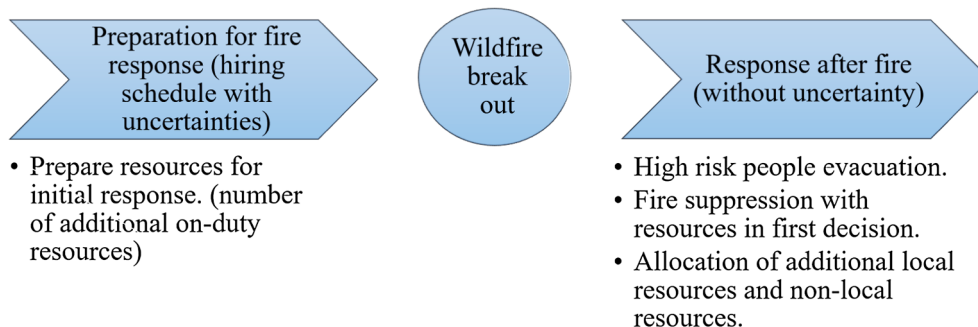


Fig. 1. Wildfire containment problem as a two-stage process.

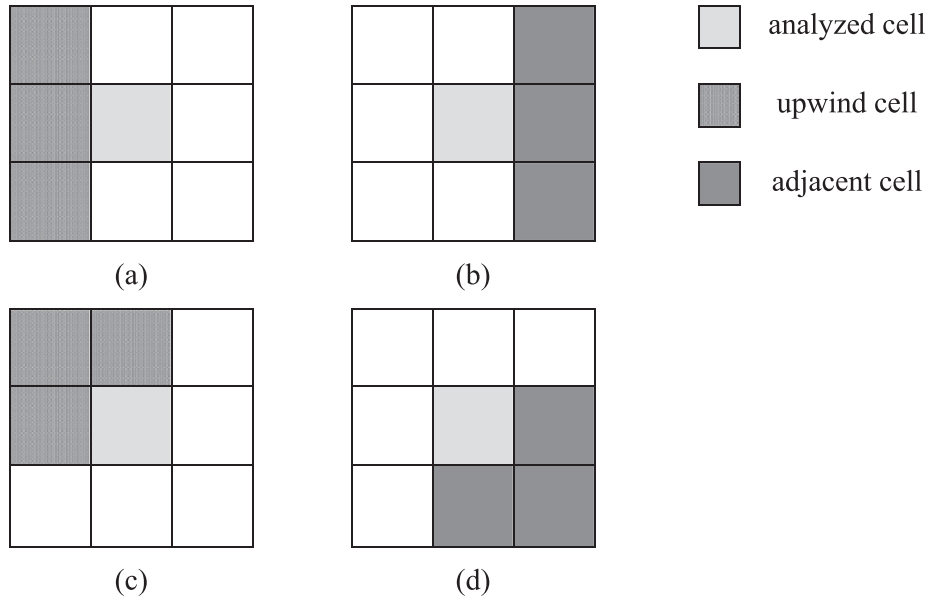


Fig. 2. Cell definitions that reflect fire-spread in different cases where (a) upwind cells in west wind scenario (b) adjacent cells in west wind scenario (c) upwind cells in northwest wind scenario (d) adjacent cells in northwest wind scenario.

There are three prerequisites for a cell to be a new high-risk cell: it should have at least one upwind burnt cell currently, it should not have been a burnt cell before, and it should have never been treated before. It is assumed that fire in a specific cell always spreads from upwind cells and can only spread to adjacent cells. The definition of upwind cells and adjacent cells are determined based on wind direction. A cell may have different adjacent cells and upwind cells in different wind directions. For example, Fig. 2(a) and (b) show that if the wind comes from the west, a cell (analyzed cell shaded in gray) has three adjacent cells on the east side and three upwind cells on the west side so that the fire can spread to this cell from west side cells and possibly spread to east side cells in the next period. Fig. 2(c) and (d) show that if the wind comes from the northwest, a cell (analyzed cell shaded in gray) has three adjacent cells on the southeast side and three upwind cells on the northwest side so that the fire can spread to this cell from northwest side cells and possibly spread to southeast side cells in the next period.

Average fire-spread rates and resources' suppression rates for different types of fuel are found in the National Wildfire Coordinating Group (NWCG) Fireline Handbook (IOSWT, 2004). The published suppression rates consider firefighters' rest time. Thus, in this model, it is assumed that all of the allocated resources will keep working at given rates until the fire is fully contained, and no specific rest time is explicitly modeled.

The model assumes that once people are evacuated, they will not go back until the fire ends. Thus, each person will only be counted once when calculating the total number of people at risk. According to the study from Paveglio, Prato, Dalenberg, and Venn, evacuation percentages differ in distinct fire sizes (Paveglio, Prato, Dalenberg, & Venn, 2014). Thus, varying evacuation percentages are used in the model based on different fire sizes in order to represent the reality that fire size affects residents' evacuation choices.

2.2. Model formulation

The deterministic equivalent of the two-stage stochastic programming model is presented below, where first stage decisions involve number of additional local on-duty resources, and second stage decisions involve additional resource allocations in each fire-spread scenario. The model includes the following parameters and decision variables:

Sets:

A	Set of cells in consideration, indexed by α
I	Set of resource types, indexed by i
K	Set of all local fire stations, indexed by k
L	Set of all nonlocal resource locations, indexed by l
T	Set of discrete periods, indexed by t
Ω	Set of all fire scenarios, indexed by ω

Parameters:

B	Budget
GS_i	Group size of resource i (required personnel)
OD_{ki}	Original on-duty resource i in station k
$LIMIT_{ki}$	Capacity of resource i in station k
$PCAP$	Capacity limit of on-duty personnel
ALR_{kit}	Capacity of additional local resource i at station k in period t ($t = 1, \dots, T$)
NR_{lit}	Additional nonlocal capacity of resource i from location l in period t ($t = 1, \dots, T$)
$INIT_{\alpha}^{\omega}$	Initial high-risk condition at cell α (equal to 1 if the cell is initial high-risk area)
RS^{ω}	Minimum required suppression rate for each cell in scenarios (chains per hour)
SUP_i	Suppression rate of resource i (chains per hour)
ODC_{ki}	Cost of hiring one additional local on-duty resource i at station k
$LORC_{ki}$	Unit operation cost of local on-duty resource i from station k
LRC_{li}	Unit operation cost of local call-for-shift resource i from station k
NRC_{li}	Unit operation cost of nonlocal resource i from location l
EVA_{α}^{ω}	Evacuation percentage at cell α in scenarios
DEN_{α}^{ω}	Population density at cell α in scenarios (because of different cell sizes)
$VALUE_{\alpha}^{\omega}$	Property and land value at cell α in scenarios
p^{ω}	Probability of scenario ω
M^{ω}	Big M in scenarios
Decision variables:	
s_{ki}	Number of additional local on-duty resources that need to be hired in station k
$x_{k \alpha t}^{\omega}$	Number of local on-duty resources i sent from station k to cell α at period t in scenario ω
$y_{k \alpha t}^{\omega}$	Number of local additional resources i sent from station k to cell α at period t in scenarios
$z_{l \alpha t}^{\omega}$	Number of nonlocal resources i sent from location l to cell α arrived at period t in scenarios
$\mu_{\alpha t}^{\omega}$	equal to 1 if cell α becomes a high-risk area at period t in scenario ω , 0 otherwise
$\kappa_{\alpha t}^{\omega}$	equal to 1 if cell α becomes a fully burnt area at period t in scenario ω , 0 otherwise
$\varphi_{\alpha t}^{\omega}$	equal to 1 if cell α starts to be treated at period t in scenario ω , 0 otherwise

Objective functions:

$$\text{Min } Obj_1 = \sum_{\omega \in \Omega} p^\omega \sum_{\alpha \in A} DEN_\alpha^\omega EVA_\alpha^\omega \sum_{t \in T} \mu_{\alpha t}^\omega \quad (1)$$

$$\begin{aligned} \text{Min } Obj_2 &= \sum_{k \in K} \sum_{i \in I} ODC_{ki} S_{ki} + \sum_{\omega \in \Omega} p^\omega \\ &\left(\sum_{k \in K} \sum_{i \in I} \sum_{\alpha \in A} \sum_{t \in T} (x_{k i \alpha t}^\omega LORC_{ki} + y_{k i \alpha t}^\omega LRC_{ki}) + \right. \\ &\left. \sum_{l \in L} \sum_{i \in I} \sum_{\alpha \in A} \sum_{t \in T} z_{l i \alpha t}^\omega NRC_{li} + \sum_{\alpha \in A} VALUE_\alpha^\omega \sum_{t \in T} \kappa_{\alpha t}^\omega \right) \quad (2) \end{aligned}$$

Constraints:

$$\begin{aligned} \sum_{k \in K} \sum_{i \in I} ODC_{ki} S_{ki} + \sum_{k \in K} \sum_{i \in I} \sum_{\alpha \in A} \sum_{t \in T} (x_{k i \alpha t}^\omega LORC_{ki} + y_{k i \alpha t}^\omega LRC_{ki}) + \\ \sum_{l \in L} \sum_{i \in I} \sum_{\alpha \in A} \sum_{t \in T} z_{l i \alpha t}^\omega NRC_{li} \leq B \quad \forall \omega \in \Omega \quad (3) \end{aligned}$$

$$S_{ki} + OD_{ki} \leq LIMIT_{ki} \quad \forall k \in K, \forall i \in I \quad (4)$$

$$\sum_{i \in I} \sum_{k \in K} GS_i(S_{ki} + OD_{ki}) \leq PCAP \quad (5)$$

$$\sum_{\alpha \in A} \sum_{t \in T} x_{k i \alpha t}^\omega \leq S_{ki} + OD_{ki} \quad \forall k \in K, \forall i \in I, \forall \omega \in \Omega \quad (6)$$

$$y_{k i \alpha 0}^\omega = 0 \quad \forall k \in K, \forall i \in I, \forall \alpha \in A, \forall \omega \in \Omega \quad (7)$$

$$\begin{aligned} \sum_{\alpha \in A} \sum_{t \in (1, T)} y_{k i \alpha t}^\omega \leq LIMIT_{ki} + \sum_{i \in (1, T)} ALR_{kit} - S_{ki} - OD_{ki} \quad \forall \\ k \in K, \forall i \in I, \forall t \in (1, T), \forall \omega \in \Omega \quad (8) \end{aligned}$$

$$\sum_{\alpha \in A} \sum_{t \in (1, T)} z_{l i \alpha t}^\omega \leq \sum_{i \in (1, T)} NR_{lit} \quad \forall l \in L, \forall i \in I, \forall \omega \in \Omega, \forall t \in T \quad (9)$$

$$\begin{aligned} \sum_{i \in I} \left(\sum_{k \in K} x_{k i \alpha t}^\omega + \sum_{k \in K} y_{k i \alpha t}^\omega + \sum_{l \in L} z_{l i \alpha t}^\omega \right) SUP_i \geq \varphi_{\alpha t}^\omega RS^\omega \quad \forall \\ \alpha \in A, \forall t \in T, \forall \omega \in \Omega \quad (10) \end{aligned}$$

$$\mu_{\alpha 0}^\omega = INIT_\alpha^\omega \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (11)$$

$$\varphi_{\alpha t}^\omega \leq \mu_{\alpha t}^\omega \quad \forall \alpha \in A, \forall t \in T, \forall \omega \in \Omega \quad (12)$$

$$\kappa_{\alpha t}^\omega = \mu_{\alpha, t-1}^\omega - \varphi_{\alpha, t-1}^\omega \quad \forall \alpha \in A, \forall t \in (1, T), \forall \omega \in \Omega \quad (13)$$

$$\begin{aligned} M^\omega \left(\mu_{\alpha t}^\omega + \sum_0^t \kappa_{\alpha t}^\omega + \sum_0^{t-1} \varphi_{\alpha t}^\omega \right) \geq \sum_{upwindcells\ of\ \alpha} \kappa_{\alpha t}^\omega \quad \forall \\ \alpha \in A, \forall t \in (1, T), \forall \omega \in \Omega \quad (14) \end{aligned}$$

$$\begin{aligned} M^\omega \left(1 - \mu_{\alpha t}^\omega + \sum_0^t \kappa_{\alpha t}^\omega + \sum_0^{t-1} \varphi_{\alpha t}^\omega \right) \geq 1 - \sum_{upwindcells\ of\ \alpha} \kappa_{\alpha t}^\omega \quad \forall \\ \alpha \in A, \forall t \in (1, T), \forall \omega \in \Omega \quad (15) \end{aligned}$$

$$\begin{aligned} M^\omega \left(1 - \mu_{\alpha t}^\omega - \sum_0^t \kappa_{\alpha t}^\omega - \sum_0^{t-1} \varphi_{\alpha t}^\omega \right) \geq - \sum_{upwindcells\ of\ \alpha} \kappa_{\alpha t}^\omega \quad \forall \\ \alpha \in A, \forall t \in (1, T), \forall \omega \in \Omega \quad (16) \end{aligned}$$

$$\sum_{t \in T} \kappa_{\alpha t}^\omega \leq 1 \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (17)$$

$$\sum_{t \in T} \varphi_{\alpha t}^\omega \leq 1 \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (18)$$

$$\sum_{t \in T} \mu_{\alpha t}^\omega \leq 1 \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (19)$$

$$\sum_{t \in T} \kappa_{\alpha t}^\omega + \sum_{t \in T} \varphi_{\alpha t}^\omega \leq 1 \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (20)$$

$$\sum_{t \in T} \kappa_{\alpha t}^\omega \leq \sum_{t \in T} \mu_{\alpha t}^\omega \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (21)$$

$$\sum_{t \in T} \varphi_{\alpha t}^\omega \leq \sum_{t \in T} \mu_{\alpha t}^\omega \quad \forall \alpha \in A, \forall \omega \in \Omega \quad (22)$$

$$\sum_{\alpha \in A} \sum_{t \in T} \varphi_{\alpha t}^\omega + \sum_{\alpha \in A} \sum_{t \in T} \kappa_{\alpha t}^\omega = \sum_{\alpha \in A} \sum_{t \in T} \mu_{\alpha t}^\omega \quad \forall \omega \in \Omega \quad (23)$$

$$S_{ki} \geq 0 \quad \forall k \in K, \forall i \in I \quad (24)$$

$$x_{k i \alpha t}^\omega, y_{k i \alpha t}^\omega \geq 0 \quad \forall k \in K, \forall i \in I, \forall \alpha \in A, \forall t \in T, \forall \omega \in \Omega \quad (25)$$

$$z_{l i \alpha t}^\omega \geq 0 \quad \forall l \in L, \forall i \in I, \forall \alpha \in A, \forall t \in T, \forall \omega \in \Omega \quad (26)$$

$$\mu_{\alpha t}^\omega, \kappa_{\alpha t}^\omega, \varphi_{\alpha t}^\omega \in \{0, 1\} \quad \forall \alpha \in A, \forall t \in T, \forall \omega \in \Omega \quad (27)$$

Objective function (1) minimizes the total expected number of people at-risk who need to be evacuated from high-risk cells over all scenarios. Objective function (2) minimizes the expected total cost of hiring additional on-duty resources, allocating local and nonlocal resources in scenarios and property losses of burnt cells over all scenarios. The total cost in (2) represents a societal cost since property losses are not incurred by fire departments.

Constraint (3) is the budget constraint that limits the total operation cost of resource preparation and allocation. Constraint (4) limits the number of engine and hand-crew resources in each station. The sum of on-duty resources in each station including original on-duty resources and additional on-duty resources should not exceed each station's capacity requirement. Constraint (5) limits the number of on-duty personnel in each fire department. The total number of on-duty personnel from all the stations should not exceed the department personnel capacity requirements. Constraints (6)–(9) are availability constraints: Constraint (6) ensures that the number of resources to be sent for firefighting cannot exceed the available resource limits. Constraint (7) ensures if a local resource is not on duty, then it is not available at time 0. Constraint (8) ensures that the number of additional local resources sent to cells must be less than the local resource availability at each time t . Constraint (9) ensures that the number of nonlocal resources sent to cells must be less than the nonlocal resource availability limitations at each time t . Constraint (10) ensures that all of the resources sent to each cell are sufficient to contain the fire, which means that the suppression rate in a specific cell should be higher than the required suppression rate. Constraint (11) sets up initial high-risk condition at the beginning of fire for each cell in each scenario. Constraint (12) ensures that only high-risk cells will be treated. Constraint (13) represents the fire-spread process geographically. If a cell is a high-risk cell and is not treated in period $t-1$, then it will become a burnt cell in period t . Constraints (14)–(16) represent the process of detecting new high-risk cells. Constraint (17) ensures each cell can become a burnt cell for at most one time. Constraint (18) ensures each cell can become a treated cell at most once. Constraint (19) ensures each cell can become a high-risk cell at most once. Constraint (20) means each cell can become a burnt cell or treated cell. If a cell has been a burnt cell, it would never be treated in the future. If it has been a treated cell, it would never become a burnt cell. Constraints (21)–(23) represent the relationships among each cell condition. Constraint (21) ensures that each burnt cell needs to be a high-risk cell previously. Constraint (22) ensures that each treated cell needs to be a high-risk cell previously. Constraint (23) ensures that each high-risk cell must become a treated cell or a burnt cell at the end. Constraints (24)–(27) are non-negativity and binary limitations for decision variables.

2.3. Methodology

The solution to this multi-objective model is found by using goal programming methodology. Goal programming is used to solve problems with multiple and conflicting objectives by setting goals for

objective functions and looking for solutions to optimize deviations from these goals (Schniederjans, 1984). Goal programming is also a relatively simple method to solve practical multi-objective problems like wildfire containment problems, which require fast solutions. Several new parameters and variables are needed to solve this problem with goal programming as listed below.

Parameters:	
G_1	Goal value of the first objective function
G_2	Goal value of the second objective function
P_1	Priority, the importance of the first objective function
P_2	Priority, the importance of the second objective function
Variables:	
d_1^-	Deviation of the first objective function below from G_1
d_1^+	Deviation of the first objective function above from G_1
d_2^-	Deviation of the second objective function below from G_2
d_2^+	Deviation of the second objective function above from G_2

Then the new model becomes the following:

$$\min\{P_1 d_1^+ + P_2 d_2^+\} \quad (28)$$

subject to

$$Obj_1 + d_1^- - d_1^+ = G_1 \quad (29)$$

$$Obj_2 + d_2^- - d_2^+ = G_2 \quad (30)$$

Constraints (3)–(27).

In this model, two objective functions are measured in different units, and the total cost is almost always significantly higher compared to the number of residents at-risk. When these two objective functions are summed up directly, the final solution may have a significant bias. To avoid this issue, the percentage normalization technique is applied here. The percentage normalization method is a simple way to ensure that each objective function can have roughly the same magnitude (Tamiz, Jones, & Romero, 1998). A normalization constant, N_j , is needed in this method. The normalization constant is defined as Eq. (31):

$$N_j = G_j/100 \quad j = 1, 2 \quad (31)$$

Then the updated model is as follows:

$$\min\left\{P_1 \frac{d_1^+}{N_1} + P_2 \frac{d_2^+}{N_2}\right\} \quad (32)$$

$$Obj_1 + d_1^- - d_1^+ = G_1 \quad (33)$$

$$Obj_2 + d_2^- - d_2^+ = G_2 \quad (34)$$

Constraints (3)–(27)

In this problem, an assumption is made that the first objective function has a much higher priority than the second objective function, i.e., $P_1 \gg P_2$, since human safety is considered to be more important than property safety in reality. In these kinds of problems with ranked goals, preemptive goal programming method is a useful tool (Winston, 1993). In preemptive goal programming, the most important objective function(s) is (are) forced to be equal to their optimal values from solving single objective function problem and then the model optimizes the deviations for less important objective functions subsequently. In this problem, V_j , $j = 1, 2$ is introduced to express the optimal values of each objective function.

Then the model is updated as follows:

$$\min d_2^+ \quad (35)$$

$$Obj_1 + d_1^- - d_1^+ = G_1 \quad (36)$$

$$Obj_2 + d_2^- - d_2^+ = G_2 \quad (37)$$

$$d_1^+ = V_1 \quad (38)$$

Constraints (3)–(27).

In the following section, we demonstrate how this model can be used in a specific area.

3. Case study

In this section, a case study is presented which considers potential fire scenarios in Santa Clara County, California. Relevant information and data are gathered from the official documents from CAL FIRE, Santa Clara County Fire Department (SCCFD), NWCG, National Wildfire Suppression Association (NWSA), United States Department of Labor (USDL), as well as Google Maps, Zillow Real Estate, and Weather Underground. Details about how data are gathered and used in this case study, along with the assumptions made are explained below.

3.1. Assumptions and data collection

In most cases, fire departments fight WUI fires without consideration of the budget if a fire event has already occurred. In this model, the budget of fire containment is set as the excess of revenues over expenditures of SCCFD in 2017, which is \$11,501,453 according to the SCCFD, 2017 Annual Report (SCCFD, 2017).

Fire-spread rate is affected by fuel model, slope, fuel moisture, and wind speed (IOSWT, 2004). A study by Jin et al. (2015) has shown that typically two fire seasons exist in California and the Santa Ana winds from October to April usually produce the more disastrous fires (Jin et al., 2015). Therefore, in this case, historical weather data from October are used. Based on the reference table from the NWCG (NWCG, 2018), the slope map of Santa Clara County (Graham & Pike, 1998) and the fuel models map of California (CAL FIRE, 2018b), that the area considered in this case study is assumed to have the average characteristics of Santa Clara homogeneously. The humidity level of analyzed area is also considered in this case study. Fig. 3 shows how the historical wind strength level and humidity in October (The Weather Company, 2018) contribute to different scenarios.

Each fire-spread scenario considers wind direction, wind speed, and humidity. The probabilities of these scenarios are calculated as in Eq. (39), in which data are collected from historical wind data and humidity data in October from 2008 to 2017 (The Weather Company, 2018).

$$P(\text{Scenario}) = \frac{\text{Number of days with wind from a specific direction with speed in a specific range and humidity in a specific range}}{\text{Total number of days in consideration}} \quad (39)$$

For each wind and humidity range, there is a corresponding fire-spread speed level according to NWCG Fireline Handbook (IOSWT, 2004). Probabilities of scenarios with different fire-spread direction and fire speed (chains per hour) are shown in Table 1. Required suppression rates in these six different fire-spread speeds are assumed to have six levels which are 2 chains per hour, 4 chains per hour, 16 chains per hour, 36 chains per hour, 64 chains per hour, 144 chains per hour, and 324 chains per hour, respectively.

We assume that each period has a length of 6 h. Accordingly, the size of each cell in the specific scenario is determined as the distance that fire can spread in 6 h without treatment so that fire-spread speed in scenarios can always be expressed as “one cell per period.” For example, when the fire-spread rate is four chains per hour, the cell size will be 24 chains.

Two types of resources are considered in the case study, engine group and hand crew. Since either engine group or hand crew can have

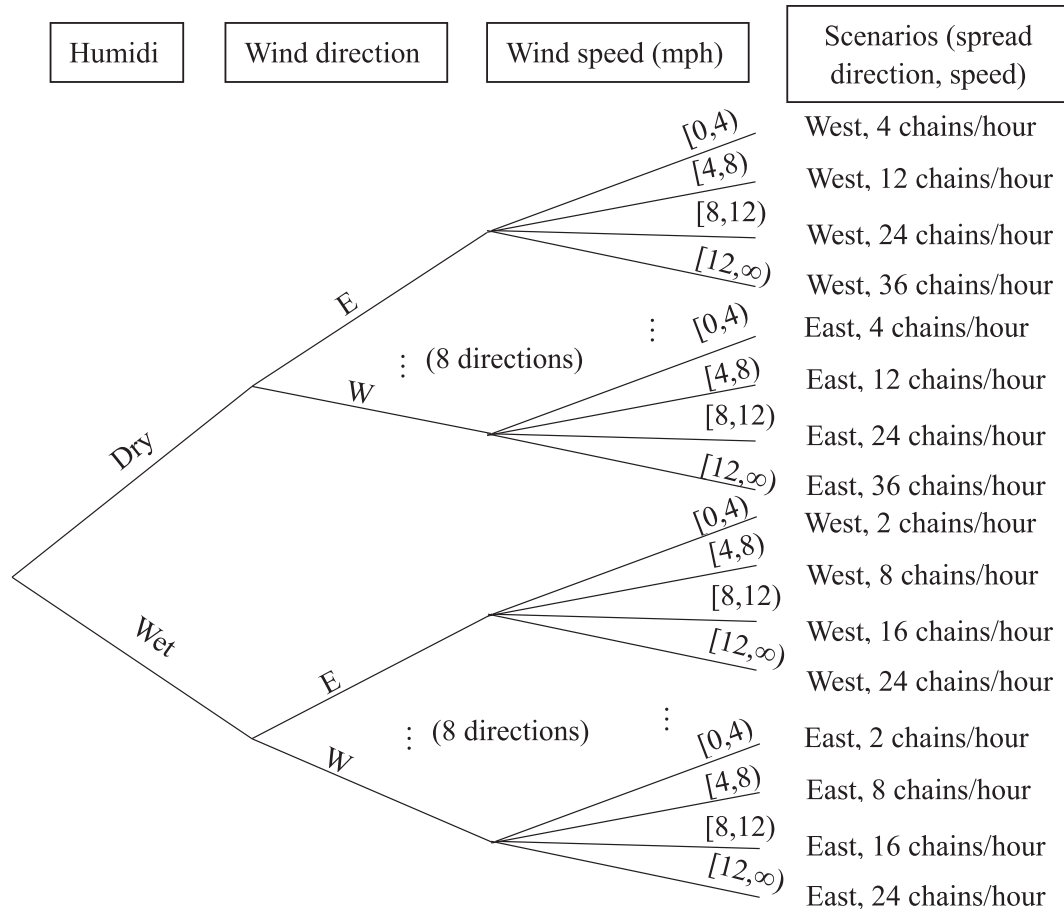


Fig. 3. Scenarios generated by humidity and wind.

different numbers of personnel, which contribute to different suppression speeds, this case study assumes that engine groups may include 1 firefighter, 2 firefighters, 3 firefighters, 4 firefighters or 5 firefighters and hand crews may include 7 firefighters or 20 firefighters based on NWCG Fireline Handbook (IOSWT, 2004). For hardwood fuel, line production rates of both resources are estimated by NWCG (IOSWT, 2004). Detailed suppression rates of different resources are shown in Table A.1.

In this example, it is assumed that only a single fire starts at period 0, and all of the operational costs are calculated based on this single fire. Resources from the local fire department are considered as local resources. Other resources including in-state resources and out-of-state resources are treated as nonlocal resources. Costs of containment of the fire include payment for additional on-duty firefighters, on-duty resources operation cost, local call-for-shift resources operation cost and nonlocal resource operation cost. The set-up cost of an additional on-

duty firefighter is assumed as \$5500 based on the data from the USDL (USDL, 2018). The set-up cost of an additional engine or truck, including preventative maintenance, training and some other indirect costs of fires, is also assumed for each station based on the data from the NWSA (NWSA, 2013). Details of engine resource set-up costs are shown in Table A.2.

Operation costs of on-duty resources, local call-for-shift resources, and nonlocal resources are estimated based on the data from the NWSA (NWSA, 2013), the Pacific Northwest Wildfire Coordinating Group (Pacific Northwest Wildfire Coordinating Group, 2017) and Google Maps (Google, 2018). Details of different distances from local stations to fire are shown in Table A.3. Since the distances of nonlocal resources are varied from station to station, average distances are assumed for neighbor-county resources, in-state resources, and out-of-state resources, respectively. Details of different nonlocal resources distances are shown in Table A.4. The equation of calculating the operational cost

Table 1
Probabilities of scenarios.

Wind direction	Fire speed (chains/hour)						
	2	4	8	12	16	24	36
E	0	0	0.003226	0	0	0	0
NE	0	0.006452	0	0.006452	0	0	0
SE	0.003226	0	0	0	0	0.003226	0
N	0	0.151613	0.029032	0.448387	0.003226	0.019355	0.009677
NW	0	0.003226	0	0.016129	0	0.003226	0
S	0	0.003226	0.035484	0.058065	0.016129	0.041935	0
SW	0	0.006452	0.006452	0.045161	0.003226	0.006452	0
W	0	0.009677	0.003226	0.054839	0	0.003226	0

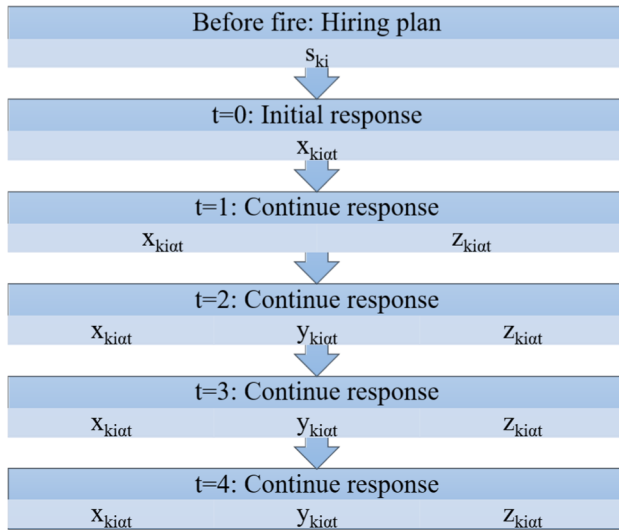


Fig. 4. Fire preparation and suppression process in the case of Santa Clara County.

to a specific resource is shown in Eq. (36)

$$\text{Operational cost} = \text{unit traffic cost} \times \text{distance} + \text{personnel payment} \times \text{number of people} \quad (40)$$

where the unit traffic cost is assumed as \$270 per mile, and personnel payment for both call-for-shift local resources and neighbor-county resources are assumed as \$616. Personnel payment for both in-state resources and out-of-state resources are assumed as \$776 based on the data from the NWSA (NWSA, 2013). Details of operational costs are shown in Tables A.5 and A.6.

The decision-making process is shown in Fig. 4. The first decision before the fire involves hiring of additional on-duty engine groups and the group size. The capacity of engine resource for each fire station in Santa Clara County is determined according to the information from the SCCFD (SCCFD, 2015). Details of engine capacities in each local fire station are shown in Table A.7. In addition to the engine capacities, there is an upper limit of 91 positions for daily duty shift in total for the county fire department (SCCFD, 2015). In reality, the number of on-duty resources is also determined by many factors such as urban fires, traffic accident rescue in addition to wildfires. Thus, in this case, an assumption is made that each station has one 3-person engine group on duty due to other considerations.

After a fire breaks out, it is assumed that engine groups from nearby state fire stations can join containment as early as the second period (after 6 h), and engine groups from neighbor departments can join containment as early as the third period (after 12 h). Local hand crews can also come as early as the third period (after 12 h). Since station 15 and the headquarters of SCCFD have the same location, additional local hand crew resources are assumed to be sent from station 15. More in-state resources can start to contain the fire as early as the fourth period (after 18 h). Additional out-of-state resources will come after 24 h. Table A.8 shows the detailed additional resource availability. Detailed resource availability is estimated based on a report from the SCCFD (2017) and CAL FIRE (2004). Response times and capacities of resources can be adjusted with more details if a higher accuracy level is required.

Fig. 5 shows the geographic areas, which are covered in the slow fire and the extremely fast fire scenarios as two different examples. Geographic information is gathered from Google Maps (Google, 2018). Population and property values for each cell are shown in Fig. 6. The value in the first row of each cell represents the number of residents in specific cell, and the value in the second row of each cell represents the land values of a specific cell. Population densities and property values

are estimated based on the cell size and publicly available information such as census data and median housing values. Population densities and property values of wildlands are collected from the result of the census (U.S. Department of Commerce, 2018) and Zillow (Zillow, 2018).

Varying evacuation percentages are assumed for different fire scenarios. Evacuation percentages for each cell are assumed as 70%, 74%, 78%, 82%, 86%, 90%, 94% for fires with different speeds respectively based on Pavaglio, Prato, Dalenberg, and Venn's study (Pavaglio et al., 2014). These estimates may be affected by the average population age in each cell and some other factors however no stratified data are currently available.

Historical data of fire locations from CAL FIRE show that some of the locations always have a higher fire risk. For example, the October 2017 Tubbs Fire happened at the same place as the September 1964 C. Hanley Fire, and the two fires also had very similar sizes. Therefore, an assumption is made that the fires in different scenarios all start from a specific point in this case study, and it is assumed that a single fire in the center of an area with 64 cells, i.e., the junction of cell 27, 28, 35, 36, starts at the beginning of the first period. However, this model could also be used for cases with multiple fires initiating at different locations at different time points by adjusting the parameters (i.e. cell conditions) related to initial high-risk cells.

3.2. Optimal solution

The model is solved by Gurobi Optimizer 7.5.1 with Anaconda Python 3.6 interface on a Dell XPS 13 – 9350 machine with Intel Core 6th Generation i5-6200U Processor, 8 GB RAM (Gurobi Optimization, 2018). The formulation in this case study included 4,318,825 decision variables and 2,933,435 constraints.

3.2.1. Single objective models

There are two single-objective models if either number of residents in danger or total cost is set as objective function. The optimal solution to a single-objective model of minimizing residents at risk is found as $G_1^* = 732.479$. The optimal solution to a single-objective model of minimizing total cost and property loss is found as $G_2^* = 49.623\text{million}$.

3.2.2. Two-objective model solved by goal programming with set priorities

When both objective functions are considered, the optimal solution to this case study is found using the goal programming technique. The goal values for each objective function are determined by solving two single objective models separately. Then the goals for the first objective function and the second objective function are set as G_1^* and G_2^* , respectively. In this case, priority is set as 0.9999 for the first objective function and 0.0001 for the second objective function after the normalization of both objective functions, since residents' safety always has a higher priority than properties in real life.

Table 2 and Fig. 7 show the optimal solution which considers both objective functions considering 56 scenarios with varying wind directions and fire speed. Only two extreme fire speed scenarios in dry weather are presented for comparison. Scenario 8 has a fire-spread speed of 4 chains per hour with northwest wind and scenario 48 has a fire-spread speed of 36 chains per hour with northwest wind.

Resource preparation and allocation results of this case are shown in Table 2. The fire in this case study involves high population density areas in several scenarios, which have a high loss potential. Thus, initial decisions are made to hire more on-duty firefighters. 4-person engine groups need to be hired in Stations 2, 6, 7, and 5-person engine groups need to be hired in Stations 0, 9, 11, and 13. The second stage decisions include resource allocation plans for each scenario after a fire breaks out. In scenario 8, which has a slow fire-spread rate, the fire is found to be fully contained with three local on-duty resources that are allocated in period 0. In scenario 48, which has rapid-fire, all the local on-duty resources need to be sent to cell 28 in period 0 due to the higher

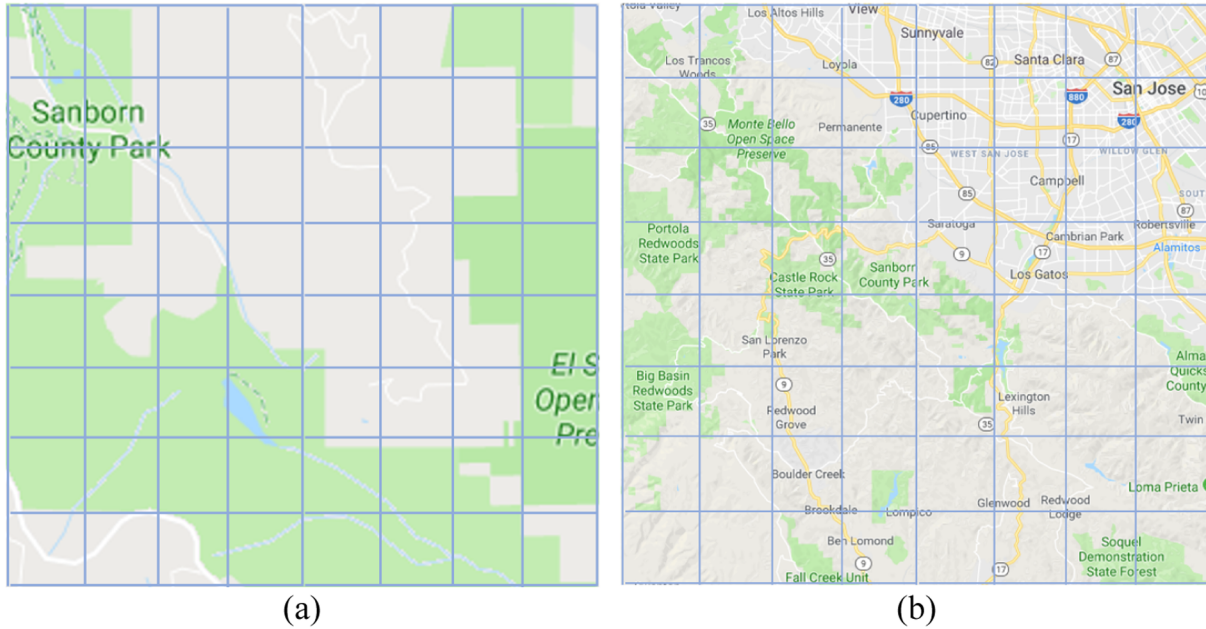


Fig. 5. Geographic areas expressed by cells (a) the fire spread 4 chains per hour and cell size is 24 chains (b) the fire spread 36 chains per hour and cell size is 216 chains.

population density and land value. However, these local on-duty resources are found to be not enough to cover all high-risk cells. During period 1, period 2 and period 3, the fire keeps expanding, and more local resources and nonlocal resources need to be assigned to high-risk areas. In period 3, the fire is going to be under control with the efforts of both local and nonlocal resources.

Fig. 7 shows cell condition results only for the specific areas of the 64 cells that are involved in these two fire-spread scenarios. In scenario 8, the weak wind coming from northwest direction leads to a slow fire which spreads to southeast direction. The cell size is 24 chains, and initial high-risk cells are determined as cells 28, 35, and 36. On-duty resources are selected and sent to these three initial high-risk cells. No more resources are needed in following periods to contain the fire. In scenario 48, the extremely strong wind coming from northwest direction leads to a fast fire which spreads to southeast direction. Initial high-risk cells are determined as 28, 35 and 36. However, compared to the cell size in scenario 8, cells in scenario 48 are much larger. The cell size in scenarios 48 is 216 chains due to “one cell per period” principle explained before. In period 0, on-duty resources are found to be

insufficient for full containment, and they are sent to cell 28, which has the highest number of residents and the highest land value. In period 1, cell 35 and cell 36 become burnt cells, and new high-risk cells are determined. Like period 0, available resources are sent to the most urgent cell, which is cell 37. In following periods, the fire keeps spreading and more resources join suppression. By period 3, resources are found to be sufficient to treat all the high-risk cells due to the arrival of a large number of nonlocal resources.

The optimal solution for this case study is achieved when $Obj1 = 732.479$ and $Obj2 = \$54.144$ million. The optimal solution shows that the first objective function has fully achieved its goal.

To check the Pareto efficiency of this solution, an integer goal programming Pareto efficiency detection technique developed by Tamiz, Mirrazavi, and Jones is applied (Tamiz, Mirrazavi, & Jones, 1998). Based on their work, the new achievement function (37) is solved.

$$\max \left\{ P_1 \frac{d_1^-}{N_1} + P_2 \frac{d_2^-}{N_2} \right\} \quad (41)$$

2	0	0	0	31	0	0	0
2.9	0.3	0.3	0.3	57.8	0.3	0.3	0.3
0	2	0	0	36	2	2	2
0.3	7.6	0.3	0.3	65.5	5.3	5.2	4.4
0	0	18	0	0	2	0	0
0.3	0.3	30	0.3	1	4.4	0.3	0.3
0	0	16	2	0	7	0	0
0.3	0.3	26.8	2.6	1	21.7	0.3	0.3
0	5	0	2	0	16	0	0
0.3	9.8	0.9	2.7	1	19.8	0.3	0.3
0	0	0	0	5	13	18	0
0.5	0.5	0.9	0.9	7.7	16	17.8	0.3
0	0	0	0	0	0	9	2
0.5	0.5	0.7	0.7	0.7	0.7	7.2	0.4
35	5	7	0	0	0	7	2
28.8	4.6	5.8	0.3	0.3	0.3	9.6	5.7

(a)

7630	13643	13702	43830	49601	48602	42549	41684
7215	13476	15224	29110	23674	19833	15812	15238
2543	15260	2760	38907	41595	45143	41900	41684
2405	14430	2646	27332	24012	17535	15382	15238
954	1272	636	28909	23871	46485	44084	41684
902	1203	601	20917	17686	19080	17159	15238
102	205	307	255	12312	20062	36279	41684
89	178	267	2587	11930	14917	15158	15238
8	15	23	30	2998	10031	12734	20842
22	45	67	89	2261	7459	7499	7619
19	67	539	530	1019	2038	1359	699
13	28	86	199	521	1041	694	347
39	135	2734	533	487	509	339	170
17	56	453	100	92	260	174	87
38	113	315	2548	2659	1172	170	85
13	33	76	515	650	295	87	43

(b)

Fig. 6. Population and property values (million) in each cell (a) slow fire scenario, low population density (b) extremely fast fire scenario, high population density.

Table 2
Resource preparation and allocation results in two selected scenarios.

First stage decision	$sk_{0i4} = 2$ $sk_{2i3} = 1$ $sk_{6i3} = 1$ $sk_{7i3} = 1$	$sk_{9i4} = 1$ $sk_{11i4} = 1$ $sk_{13i4} = 1$		
Second stage decision of scenario 8 (Northwest wind lead to 4 chains per hour fire-spread speed)	t_0 $x_{k2i2\alpha28t0}^{\omega8} = 1$ $x_{k7i2\alpha36t0}^{\omega8} = 1$ $x_{k7i3\alpha35t0}^{\omega8} = 1$	t_1	t_2	t_3
Second stage decision of scenario 48 (Northwest wind lead to 36 chains per hour fire-spread speed)	t_0 $x_{k0i2\alpha28t0}^{\omega48} = 1$ $x_{k0i4\alpha28t0}^{\omega48} = 2$ $x_{k1i2\alpha28t0}^{\omega48} = 1$ $x_{k2i2\alpha28t0}^{\omega48} = 1$ $x_{k2i3\alpha28t0}^{\omega48} = 1$ $x_{k3i2\alpha28t0}^{\omega48} = 1$ $x_{k4i2\alpha28t0}^{\omega48} = 1$ $x_{k5i2\alpha28t0}^{\omega48} = 1$ $x_{k6i2\alpha28t0}^{\omega48} = 1$ $x_{k6i3\alpha28t0}^{\omega48} = 1$ $x_{k7i2\alpha28t0}^{\omega48} = 1$ $x_{k7i3\alpha28t0}^{\omega48} = 1$ $x_{k8i2\alpha28t0}^{\omega48} = 1$ $x_{k9i2\alpha28t0}^{\omega48} = 1$ $x_{k9i4\alpha28t0}^{\omega48} = 1$ $x_{k10i2\alpha28t0}^{\omega48} = 1$ $x_{k11i2\alpha28t0}^{\omega48} = 1$ $x_{k11i4\alpha28t0}^{\omega48} = 1$ $x_{k12i2\alpha28t0}^{\omega48} = 1$ $x_{k13i2\alpha28t0}^{\omega48} = 1$ $x_{k13i4\alpha28t0}^{\omega48} = 1$	t_1 $y_{k4i4\alpha37t1}^{\omega48} = 2$ $y_{k5i4\alpha37t1}^{\omega48} = 1$ $y_{k14i5\alpha37t1}^{\omega48} = 6$ $z_{i0i4\alpha37t1}^{\omega48} = 8$	t_2 $y_{k14i5\alpha51t2}^{\omega48} = 2$ $z_{i1i2\alpha53t2}^{\omega48} = 18$ $z_{i1i3\alpha52t2}^{\omega48} = 18$ $z_{i1i4\alpha46t2}^{\omega48} = 15$ $z_{i1i4\alpha51t2}^{\omega48} = 14$ $z_{i1i4\alpha53t2}^{\omega48} = 5$	t_3 $z_{i2i5\alpha55t3}^{\omega48} = 24$ $z_{i2i5\alpha63t3}^{\omega48} = 24$ $z_{i2i6\alpha62t3}^{\omega48} = 9$
The result of objective functions	Obj1 = 732.479 (Number of residents at risk) Obj2 = \$54.1442 $\times 10^6$ (Total cost of preparation and fire containment plus property losses)			

The new model is infeasible, i.e. no point is found for this new achievement function, which proves that no other solution dominating the current optimal solution is found in the original feasible region. Thus, the solution obtained above is Pareto efficient.

3.2.3. Two-objective model solved by preemptive goal programming

Like the basic Goal Programming method, Preemptive Goal Programming also requires for the setting of goals to each objective function. Similarly, the goals for the first objective function and the second objective function are set as G_1^* and G_2^* , respectively. Since residents' safety is always more important than properties in real life, $P_1 \gg P_2$ is assumed in this case. Then, the model will be solved with preemptive goal programming method by including a constraint that sets the deviation from first objective G_1^* to be 0.

The solution produced by preemptive goal programming method shows that the minimum number of residents at risk is 732.479 and the minimum total cost from such fire is \$54.145 million.

Comparing results from both basic goal programming with priorities of 0.9999 and 0.0001 and preemptive goal programming, we did not observe a significant difference in optimal solutions. Both methods produced solutions that achieve the objective of minimizing number of residents at risk and exceeded the goal of total cost for roughly 54 million. However, the total runtime of finding solution with preemptive goal programming is more than three hours, which is much longer compared to the total runtime of 180.89 s with basic goal programming.

3.3. Numerical experiments

In this section, sensitivity analysis results by changing priorities, goals, on-duty personnel capacity, or call-for-shift personnel capacity are presented. In these numerical experiments, objective functions have been normalized in order to evaluate the impact of P_1 and P_2 . P_1 and P_2 are chosen between 0 and 1 to show the importance of both objective functions. Optimal objective function values of the individual problems with single objectives are the best values that the model can achieve. Thus, goal values are initially set as the solutions of single objective models with corresponding parameters. When priorities are fixed and goal values are varied, the results show the sensitivity to goal values. When goal values are fixed and priorities are varied, the results show the sensitivity to priorities. When capacity value is varied, the results show its sensitivity to corresponding capacity.

Table 3 shows the experiments results with $G_1 = G_1^*$, $G_2 = G_2^*$. These experiments have the same goal value while the priority factors are varied. The result of preemptive goal programming experiment is also shown in Table 3. Changes in percentage of objective function values refer to the objective value changes comparing to G_1^* and G_2^* . In this set of experiments, as the first objective function becomes less important, its solution becomes worse while the second objective function's solution keep improving until the goal value \$49.623 million is fully achieved, as expected.

Table 4 shows the results of the experiments with $P_1 = 0.9999$, $P_2 = 0.0001$ with varying goal values. Percent changes in objective function values refer to the changes compared to G_1^* and G_2^* . In this set

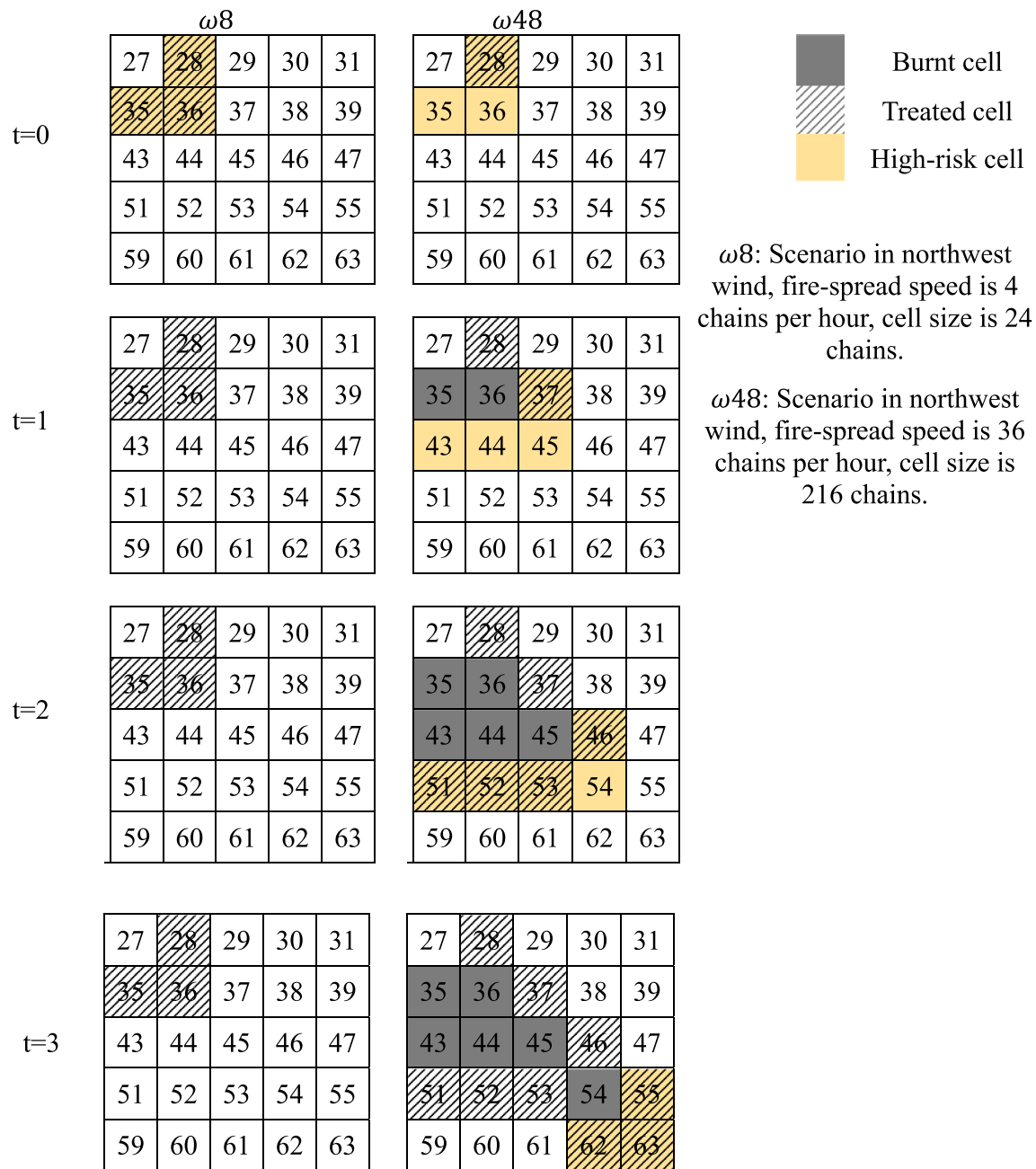


Fig. 7. Cell conditions in two selected scenarios.

Table 3

Numerical Experiments Result with Varied Priorities ($G_1 = 732.479$, $G_2 = 49.623M$).

Experiment No.	P_1	P_2	Obj_1	% change in Obj_1	Obj_2 (\$)	% change in Obj_2	Dev1	Dev2 (\$)	Time (s)
1	Preemptive GP		732.479	0.00%	54.145M	9.11%	0	4.523M	10863.77
2	0.9999	0.0001	732.479	0.00%	54.144M	9.11%	0	4.522M	180.89
3	0.999	0.001	732.479	0.00%	54.128M	9.08%	0	4.505M	483.94
4	0.99	0.01	732.824	0.05%	51.331M	3.44%	0.346	1.708M	993.52
5	0.9	0.1	732.824	0.05%	51.330M	3.44%	0.346	1.707M	1626.90
6	0.5	0.5	*	*	*	*	*	*	*
7	0.1	0.9	767.668	4.80%	49.623M	0.00%	35.189	0	201.41
8	0.01	0.99	767.668	4.80%	49.623M	0.00%	35.189	0	355.97
9	0.001	0.999	767.668	4.80%	49.623M	0.00%	35.189	0	859.79
10	0.0001	0.9999	767.668	4.80%	49.623M	0.00%	35.189	0	210.60

Note: Experiment 2 is the one with original parameters.

* Optimum solution could not be reached in 20 h.

Table 4Numerical experiments with varying goal values ($P_1 = 0.9999$, $P_2 = 0.0001$).

Exp No.	Change in goals				Change in optimal objective values				Deviations		Time (s)
	G_1	% change in G_1	G_2 (\$)	% Change in G_2	Obj_1	% change in Obj_1	Obj_2 (\$)	% change in Obj_2	Dev1	Dev2 (\$)	
1	732.479	0%	49.623M	0%	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
2	732.479	0%	54.585M	10%	732.479	0.00%	54.225M	9.28%	0.000	-0.36M	168.95
3	732.479	0%	59.547M	20%	732.479	0.00%	55.459M	11.76%	0.000	-4.088M	164.60
4	732.479	0%	74.434M	50%	732.479	0.00%	56.324M	13.50%	0.000	-18.11M	161.73
5	805.727	10%	49.623M	0%	767.668	4.80%	49.628M	0.01%	-38.059	0.006M	263.56
6	805.727	10%	54.585M	10%	788.584	7.66%	53.552M	7.92%	-17.143	-1.033M	161.50
7	805.727	10%	59.547M	20%	776.905	6.07%	58.051M	16.99%	-28.822	-1.496M	150.48
8	805.727	10%	74.434M	50%	801.331	9.40%	72.929M	46.97%	-4.396	-1.505M	143.66
9	878.975	20%	49.623M	0%	767.668	4.80%	49.630M	0.02%	-111.307	0.007M	217.91
10	878.975	20%	54.585M	10%	798.162	8.97%	52.694M	6.19%	-80.813	-1.891M	143.86
11	878.975	20%	59.547M	20%	799.987	9.22%	59.248M	19.40%	-78.988	-0.299M	151.23
12	878.975	20%	74.434M	50%	751.254	2.56%	66.525M	34.06%	-127.721	-7.909M	148.40
13	1098.718	50%	49.623M	0%	767.668	4.80%	49.641M	0.04%	-331.050	0.018M	193.76
14	1098.718	50%	54.585M	10%	744.002	1.57%	54.584M	10.00%	-354.716	-0.001M	152.56
15	1098.718	50%	59.547M	20%	780.697	6.58%	58.567M	18.02%	-318.021	-0.98M	152.75
16	1098.718	50%	74.434M	50%	878.302	19.91%	73.673M	48.47%	-220.416	-0.761M	146.20

Note: Experiment 1 is the one with original parameters.

of experiments, the first objective function which minimizes residents at-risk is fully achieved when the first goal value is set as G_1^* . Similarly, the second objective function that minimizing total cost is closer to the goal when the second goal value is set as G_2^* compared to other higher values. When either of objective functions is relaxed by related high goal value, the solution to corresponding objective function will become worse. When both objective functions are relaxed to higher goal values, optimal values of both objective functions become much worse.

Table 5 shows the results of experiments with varying on-duty personnel capacities. In this set of experiments, as on-duty personnel capacity changes, corresponding goal values also change based on the new solutions of single objective problems. Percent changes in objective function values refer to the objective value changes comparing to G_1^* and G_2^* . Results show that increasing on on-duty personnel capacity leads to a very small improvement in the second objective function value as the first objective function has achieved the goal. Ten percent decrease on on-duty personnel capacity didn't affect the optimal objective values obviously. However, bigger decreases in capacity worsen the objective function values significantly. The results suggest that decision makers need to pay more attention on evaluating overall losses when they plan to cut down capacity of on-duty personnel.

Table 6 shows the experiments results with varied call-for-shift personnel capacities. Experiment 1 is the one with original parameters. Changes in percentage of objective function values refer to the objective value changes comparing to G_1^* and G_2^* . In this set of experiments, as call-for-shift personnel capacity changes, correspond goal values also change. Results in Table 6 show that increases on call-for-shift personnel capacity lead to improvements on both objective functions comparing to the original optimal values of each objective function. Small decreases that are equal to or less than twenty percent don't lead to significant increase on objective values. However, any decrease

beyond fifty percent on call-for-shift personnel capacity causes both objective function values to increase significantly. These results suggest that decision makers may consider increasing the capacity of call-for-shift personnel to make both potential at-risk residents and total cost of wildfire lower.

Table 7 shows the experiments results with varied nonlocal engine capacities. Changes in percentage of objective function values refer to the objective value changes comparing to G_1^* and G_2^* . In this set of experiments, changes of engine capacities in different nonlocal resource types show different influence levels. Results in Table 7 show that increases on engine capacities of neighbor resources and in-state resources lead to improvements on both objective functions comparing to the original optimal values of each objective function, while decreases on engine capacities of neighbor resources and in-state resources lead to even more significant deteriorations. However, changes of engine capacity in out-of-state resources don't cause obviously changes to objective functions. Since nonlocal resources are not hired by local fire department, they will join fire containment only if cooperation contracts were signed before fire. Thus, increasing of nonlocal resource capacity doesn't produce additional hiring cost. When the related nonlocal capacity constraint is tight, increasing of capacity will lead to cost savings on property losses compared to the increasing on operation cost of nonlocal resources. These results suggest that decision maker may consider increasing the capacity of both neighbor engine resource and in-state engine resource to reduce the potential at-risk residents and total cost of wildfire by expanding cooperation with other fire departments.

Fig. 8 shows the cell conditions with two ignition points in scenario 48. In this experiment, wildfire starts from two ignition points, which lead to multiple initial high-risk cells locating at two detached areas. One of the ignition points is the same as the one in Section 3.2.2, which produces

Table 5Numerical experiments result with varied on-duty personnel capacities ($P_1 = 0.9999$, $P_2 = 0.0001$).

Experiment No.	On-duty personnel capacity	% change in capacity	G_1	G_2 (\$)	Obj_1	% change in Obj_1	Obj_2 (\$)	% Change in Obj_2	Dev1	Dev2 (\$)	Runtime (s)
1	91	0	732.479	49.623M	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
2	100	10%	732.479	49.622M	732.479	0.00%	54.134M	9.09%	0.000	4.512M	294.01
3	109	20%	732.479	49.622M	732.479	0.00%	54.134M	9.09%	0.000	4.512M	233.04
4	137	50%	732.479	49.622M	732.479	0.00%	54.134M	9.09%	0.000	4.512M	234.49
5	82	-10%	732.479	49.623M	732.479	0.00%	54.146M	9.12%	0.000	4.524M	196.86
6	73	-20%	914.970	165.017M	914.970	24.91%	169.550M	241.68%	0.000	4.533M	193.02
7	46	-50%	1390.178	256.650M	1390.178	89.79%	258.195M	420.32%	0.000	1.545M	212.45

Note: Experiment 1 is the one with original parameters.

Table 6Numerical experiments result with varied call-for-shift personnel capacities ($P_1 = 0.9999$, $P_2 = 0.0001$).

Experiment No.	Call-for-shift personnel capacity	% change in capacity	G_1	G_2 (\$)	Obj_1	% Change in Obj_1	Obj_2 (\$)	% change in Obj_2	Dev1	Dev2 (\$)	Runtime (s)
1	61	0	732.4788	49.6226M	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
2	67	10%	732.4788	49.6217M	*	*	*	*	*	*	*
3	73	20%	732.3017	49.6027M	732.302	−0.02%	51.295M	3.37%	0.000	1.692M	662.85
4	92	50%	728.4357	48.7066M	728.436	−0.55%	49.591M	−0.06%	0.000	0.884M	321.31
5	55	−10%	732.4788	49.6226M	732.479	0.00%	54.142M	9.11%	0.000	4.519M	238.69
6	49	−20%	732.4788	49.6237M	732.479	0.00%	54.152M	9.13%	0.000	4.528M	184.55
7	31	−50%	848.3121	121.8069M	848.312	15.81%	126.338M	154.60%	0.000	4.531M	172.22

Note: Experiment 1 is the one with original parameters.

* Optimum solution could not be reached in 15 h.

Table 7Numerical experiments result with varied nonlocal engine capacities ($P_1 = 0.9999$, $P_2 = 0.0001$).

Experiment No.	Nonlocal engine capacity (neighbor)	% change in capacity	G_1	G_2 (\$)	Obj_1	% change in Obj_1	Obj_2 (\$)	% Change in Obj_2	Dev1	Dev2 (\$)	Runtime (s)
1.1	8	0	732.479	49.623M	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
1.2	12	50%	728.436	48.692M	728.436	−0.55%	49.585M	−0.08%	0.000	0.893M	213.81
1.3	4	−50%	848.312	121.8M	848.312	15.81%	126.362M	154.65%	0.000	4.561M	169.59
	Nonlocal engine capacity (in-state)	% change in capacity	G_1	G_2 (\$)	Obj_1	% change in Obj_1	Obj_2 (\$)	% Change in Obj_2	Dev1	Dev2 (\$)	Runtime (s)
2.1	70	0	732.479	49.623M	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
2.2	105	50%	728.613	48.735M	728.613	−0.53%	52.449M	5.70%	0.000	3.715M	191.35
2.3	35	−50%	749.498	52.275M	749.498	2.32%	63.397M	27.76%	0.000	11.122M	178.16
	Nonlocal engine capacity (out-of-state)	% change in capacity	G_1	G_2 (\$)	Obj_1	% change in Obj_1	Obj_2 (\$)	% Change in Obj_2	Dev1	Dev2 (\$)	Runtime (s)
3.1	229	0	732.479	49.623M	732.479	0.00%	54.144M	9.11%	0.000	4.522M	180.89
3.2	344	50%	732.479	49.622M	732.479	0.00%	54.139M	9.10%	0.000	4.517M	180.27
3.3	115	−50%	732.479	49.622M	732.479	0.00%	54.134M	9.09%	0.000	4.512M	223.64

Note: Experiment 1.1, 2.1, and 3.1 are the experiments with original parameters.

initial high-risk cells of cell 28, 35, 36; the other ignition point is in the cell 16, which produces only one initial high-risk cell of cell 16. The objective function values of this experiment are $Obj_1 = 787.221$ and $Obj_2 = \$87.677$ million, respectively. Compared to the original experiment in Section 3.2.2 and the original objective function values of $Obj_1 = 732.479$ and $Obj_2 = \$54.144$ million, results from this experiment show that the having multiple ignition points leads to significant increase on number of both high-risk cells and burnt cells, which as a result increases both objective function values. In Fig. 7, there are 6 burnt cells and 9 treated cells produced in total until the fully containment. However, Fig. 8 shows that there are 15 burnt cells and 12 treated cells produced in total. Even if the increased ignition point is in the cell 16, cell 46 and 53, which are far away from cell 16, also become to burnt cells in Fig. 8. These two additional burnt cells are caused by having limited resources to fight with fires in two separate areas. Since one more ignition point produces more initial high-risk cells, resource allocation decisions are made based on evaluation of the new condition, which means in the new experiment, cell 46 and 53 are less important than cell 18 and 26. Due to the presence of limited resources, the increasing of high-risk cells, burnt cells, and treated cells is unavoidable if multiple ignition points exist. This result suggests that fire prevention is another important aspect to achieve the minimization of residents at risk and total cost. Once more ignition points appear, the challenge of fire containment will be even bigger. To reduce the at-risk residents and total cost, decision makers should make efforts to reduce wildfire ignition points.

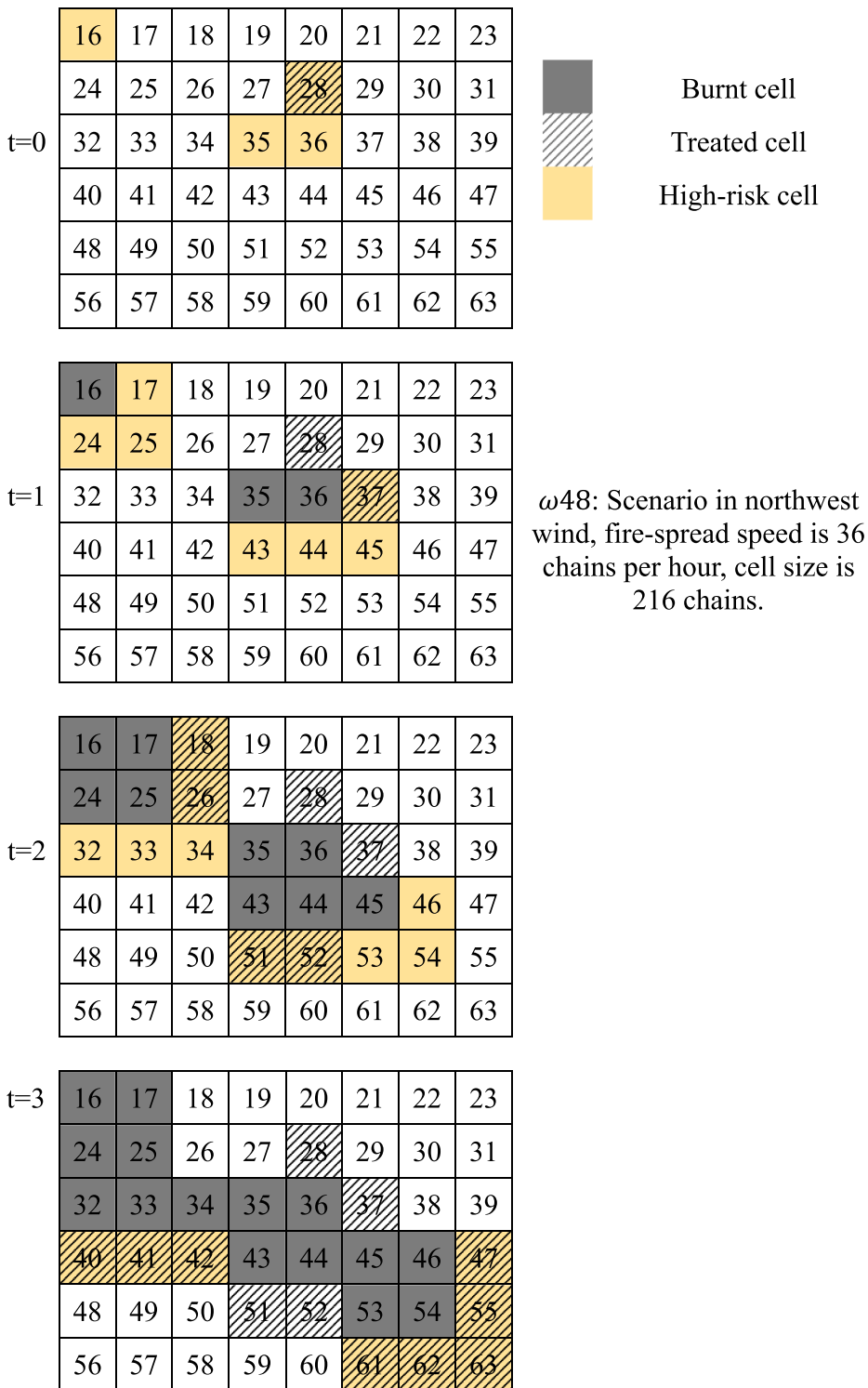
4. Conclusions

This paper presents a new integer stochastic goal programming model to solve the resident evacuation and resource allocation problem

for wildfire preparedness and response. The proposed model minimizes both the number of residents at risk and total cost of fire containment and property loss. This paper explores a new approach to keep the number of variables constant by increasing the size of grids for the scenarios, which have higher fire-spread speed. Results show that this stochastic goal programming model can produce solutions to highly comprehensive realistic WUI wildfire problems with all possible fire scenarios within a very short computation time. In each scenario, the fire-spread process is influenced by response actions along with the wind speed. Numerical experiments with different priority settings indicate that it is possible to find solutions by considering a trade-off between the two objective functions. When decision makers plan to make changes to capacity, they need to evaluate potential significant effects on at-risk residents and total cost. Experiments with multiple ignition points also point out to the importance of fire prevention efforts to reduce the probability of having multiple ignition points, as during such an event, suppression efforts would be less effective due to having limited resources to fight fires in multiple areas.

A disadvantage of the goal programming methodology that was used in this paper is that the decision maker must make assumptions to goals and their priorities, which may be difficult to determine. However, it is a relatively simple method in that only a few additional variables are required to solve multi-objective problems. The impact of the priority parameters on the optimal solution is investigated with numerical experiments. For complex real-world problems such as WUI wildfire preparedness and containment, goal programming ensures that a solution is found in a short time and is easy to understand.

In practical application of this model, the effectiveness of the produced solution is highly related to the accuracy of the collected data. However, in large-scale real-life problems, data cannot be simply



collected from published resources. For example, land values and residents in each cell are key parameters for resource allocation decisions, while these values are gathered from published data assuming a uniform distribution within a cell. A more accurate data acquisition and conversion module is needed to be used along with the fire containment model.

An important assumption in this model is that wind speed and direction do not change once a fire breaks out, which is a reasonable assumption for most of the relatively stable situations where fires are contained within a few hours or in the first few days, following the

assumption of predictable seasonal wind behavior in a known topology. But in some exceptional cases, the wind suddenly changes before fire containment or fire suppression lasts for several weeks, then fire-spread direction may change between periods. In those cases, the problem becomes a multi-stage stochastic programming problem. Thus, the formulation needs to be extended to reflect changing scenarios.

Another assumption made in this model is that once resource allocation decisions are made, all the allocated resources stay in a specific area until the fire is fully contained. In reality, if decision makers decide that the fire will not spread in their current area but keeps spreading in

other areas, firefighting resources may be reallocated.

Fuel management focusing on specific high-risk areas to reduce fire risk and magnitude is also a critical part of wildfire preparation. Thus, a more detailed model, which considers both fuel management and fire suppression, can be developed in the future. Furthermore, the model presented in this paper has the limiting assumption of homogenous geographical structure in each cell, which can only provide a framework for more realistic applications with heterogeneous geographic areas. More realistic models can be developed by incorporating a mathematical model along with a geographical information system software.

Disaster response problems including wildfire containment are problems with high complexity. The model presented here can only provide solutions in a system level by following the basic assumptions made. When dealing with real problems, wind conditions and the fire-spread direction usually behave with much more complexity than the assumptions made in this model. Resource allocation and resident evacuation are also greatly influenced by traffic conditions and timing of the fire. More detailed plans can be made by combining resource allocation models with wind prediction model, or fire behavior prediction model with high precision.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cie.2019.106101>.

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